Coding Theory for Mathematicians: Sets, Linear Algebra, and Algebraic Geometry

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1 Philosophy: codes as objects and morphisms

A block code is a subset $C \subseteq \Sigma^n$. For mathematicians, the most useful additional structures are:

- $\Sigma = \mathbb{F}_q$ a finite field and C a *linear subspace* of $\mathbb{F}_q^n$.
- C as an *image* of an encoding morphism $Enc : \Sigma^k \to \Sigma^n$.
- C as a *kernel* of linear equations (parity checks) $H : \mathbb{F}_q^n \to \mathbb{F}_q^{n-k}$.
- (AG viewpoint) C as an *evaluation image* of functions/sections on a variety/curve.

2 Basic definitions (sets, functions, metric)

2.1 Codes and encoders

Definition 2.1 (Code)**.** A code C of block length n over alphabet Σ is a subset $C \subseteq \Sigma^n$. Elements are codewords.

Definition 2.2 (Encoder)**.** An encoding map is a function $Enc : \Sigma^k \to \Sigma^n$. The code is the image $C = \mathrm{im}(Enc)$.

2.2 Errors and erasures

During transmission, a codeword $c \in C$ becomes $\tilde{c} \in \Sigma^n$.

- **Erasure:** positions of corruption are known (symbol is "?").
- **Error:** positions are unknown (symbol may flip to another value).

2.3 Hamming distance

Definition 2.3 (Hamming distance)**.** For $x, y \in \Sigma^n$,

$$\Delta(x, y) = |\{i : x_i \neq y_i\}|.$$

Definition 2.4 (Minimum distance)**.**

$$d(C) = \min_{\substack{c,c' \in C \\ c \neq c'}} \Delta(c, c').$$

Theorem 2.5 (Distance controls correction/detection)**.** *A code of distance d can:*

(a) correct up to $d-1$ erasures;

(b) detect up to $d-1$ errors;

(c) correct up to $\left\lfloor \frac{d-1}{2} \right\rfloor$ errors.

Proof idea. Hamming balls of radius $\lfloor (d-1)/2 \rfloor$ around distinct codewords are disjoint. Nearest-neighbor decoding is unambiguous inside a ball. □

3 Linear codes as subspaces, images, kernels

From now on $\Sigma = \mathbb{F}_q$.

3.1 Linear codes

Definition 3.1 (Linear code)**.** A linear $(n,k)_q$ code is a k-dimensional $\mathbb{F}_q$-subspace $C \leq \mathbb{F}_q^n$.

Definition 3.2 (Generator matrix)**.** A generator matrix $G \in \mathbb{F}_q^{k \times n}$ has row space C. Encoding is the linear map

$$Enc : \mathbb{F}_q^k \to \mathbb{F}_q^n, \quad m \mapsto mG.$$

Definition 3.3 (Parity-check matrix)**.** A parity-check matrix $H \in \mathbb{F}_q^{(n-k) \times n}$ satisfies

$$C = \{c \in \mathbb{F}_q^n : Hc^\top = 0\} = \ker(H).$$

3.2 Weight and distance in linear codes

Definition 3.4 (Hamming weight)**.** For $x \in \mathbb{F}_q^n$, $\mathrm{wt}(x) = |\{i : x_i \neq 0\}|$.

Proposition 3.5. *For a linear code C,*

$$d(C) = \min\{\mathrm{wt}(c) : c \in C,\ c \neq 0\}.$$

3.3 Syndromes

Definition 3.6 (Syndrome map)**.** Given H, define $s : \mathbb{F}_q^n \to \mathbb{F}_q^{n-k}$ by $s(y) = Hy^\top$. Then $y \in C \iff s(y) = 0$.

If $y = c + e$ with $c \in C$, then $s(y) = He^\top$. So decoding often reduces to:

$$\text{compute } s(y) \ \rightsquigarrow\ \text{infer error pattern } e.$$

4 Classic examples: parity and Hamming codes

4.1 Single parity-check code

Example 4.1 ($(n, n-1, 2)_2$ parity code)**.** Let $C = \{x \in \mathbb{F}_2^n : x_1 + \cdots + x_n = 0\}$. Then $d = 2$: detects 1 error, corrects 1 erasure.

4.2 The $(7,4,3)_2$ Hamming code

Example 4.2 (Hamming $(7,4,3)$)**.** Let columns of $H \in \mathbb{F}_2^{3\times 7}$ be all nonzero vectors in $\mathbb{F}_2^3$:

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Then $C = \ker(H)$ has parameters $(7,4,3)_2$ and corrects 1 error. Indeed, for a single-bit error e at position j, $s(y) = He^\top$ equals the j-th column of H, identifying j.

5 Bounds: Singleton and Hamming

5.1 Singleton bound

Theorem 5.1 (Singleton)**.** *Any $(n,k,d)_q$ code satisfies $d \le n-k+1$.*

Proof. Puncture (delete) $d-1$ coordinates. Distinct codewords remain distinct, so $|C| \le q^{n-(d-1)}$, i.e. $k \le n-d+1$. □

Codes meeting Singleton with equality are *MDS* (maximum distance separable).

5.2 Hamming bound

Let $V_q(n,e) = \sum_{i=0}^{e} \binom{n}{i}(q-1)^i$ be the Hamming ball volume.

Theorem 5.2 (Hamming bound)**.** *If $C \subseteq \mathbb{F}_q^n$ has distance d and $e = \lfloor (d-1)/2 \rfloor$, then*

$$|C| \cdot V_q(n,e) \le q^n.$$

Example 5.3 (Hamming code is perfect)**.** For $(7,4,3)_2$, $e = 1$, $V_2(7,1) = 1+7 = 8$, so $|C| \le 2^7/8 = 16$ and $k \le 4$. Equality holds; this is a perfect code.

6 Reed–Solomon codes: $\mathbb{P}^1$, evaluation, and MDS

6.1 Evaluation construction

Fix distinct $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$ with $n \le q$. Let $V = \{f \in \mathbb{F}_q[x] : \deg f < k\}$ (dimension k). Define

$$\mathrm{ev} : V \to \mathbb{F}_q^n, \quad f \mapsto (f(\alpha_1), \ldots, f(\alpha_n)).$$

Its image is the Reed–Solomon code $RS_q(n,k)$.

6.2 Distance

Theorem 6.1 (RS is MDS)**.** *$RS_q(n,k)$ has minimum distance $d = n-k+1$.*

Proof. If $f \ne g$ with $\deg(f-g) < k$, then $f-g$ has $\le k-1$ roots, so $(f(\alpha_i))$ and $(g(\alpha_i))$ agree in at most $k-1$ coordinates, hence differ in at least $n-(k-1)$. □

6.3 AG translation: $\mathbb{P}^1$ and line bundles

On $\mathbb{P}^1$, polynomials of degree $\leq k-1$ identify with global sections of $\mathcal{O}_{\mathbb{P}^1}(k-1)$. Choosing evaluation points is choosing a divisor $D = P_1 + \cdots + P_n$ of rational points. RS is the code $C_L(D, G)$ with $G = (k-1) \cdot \infty$ on $\mathbb{P}^1$.

7 Worked example: RS over $\mathbb{F}_7$ and decoding (Berlekamp–Welch)

We do a fully explicit encode + decode by algebra (no "closest codeword" black box).

7.1 Setup

Take $q = 7$, $n = 6$, $k = 3$. Then $d = n - k + 1 = 4$, so we can correct $t = \lfloor (d-1)/2 \rfloor = 1$ error.

Choose evaluation points $\alpha_1, \ldots, \alpha_6 = 1, 2, 3, 4, 5, 6 \in \mathbb{F}_7$. Messages are polynomials $f(x) = a + bx + cx^2$ in $\mathbb{F}_7[x]$.

7.2 Encoding example

Let

$$f(x) = 2 + 3x + x^2 \in \mathbb{F}_7[x].$$

Evaluate:

x	1	2	3	4	5	
$f(x)$	$2+3+1=6$	$2+6+4=12\equiv 5$	$2+9+2=13\equiv 6$	$2+12+2=16\equiv 2$	$2+15+4=21\equiv 0$	$2+18$

So the codeword is

$$c = (6, 5, 6, 2, 0, 0) \in \mathbb{F}_7^6.$$

7.3 One-error corruption

Suppose position $x = 4$ is corrupted: $2 \mapsto 1$. Received word:

$$y = (6, 5, 6, 1, 0, 0).$$

7.4 Berlekamp–Welch decoding (for $t = 1$)

We seek polynomials $E(x)$ and $N(x)$ such that:

$$E(\alpha_i)\, y_i = N(\alpha_i) \quad \text{for } i = 1, \ldots, 6,$$

with degree constraints

$$\deg E \leq t = 1, \qquad \deg N \leq k - 1 + t = 3.$$

If E vanishes at the error location, then $N = E \cdot f$ and we recover $f = N/E$.

Write

$$E(x) = x - \beta \quad (\beta \in \mathbb{F}_7), \qquad N(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3.$$

Then constraints become for each i:

$$(\alpha_i - \beta)\, y_i = u_0 + u_1 \alpha_i + u_2 \alpha_i^2 + u_3 \alpha_i^3.$$

This is a linear system in the unknowns $\beta, u_0, u_1, u_2, u_3$, except β appears multiplicatively. A common trick is to represent $E(x) = e_0 + e_1 x$ with unknown coefficients and solve linearly.

So set $E(x) = e_0 + e_1 x$ (not both zero), and $N(x)$ as above. Solve:

$$(e_0 + e_1\alpha_i)y_i = N(\alpha_i), \quad i = 1, \dots, 6.$$

Unknowns: $e_0, e_1, u_0, u_1, u_2, u_3$ (6 equations, 6 unknowns up to scaling of E, N). One can normalize $e_1 = 1$ (if $e_1 \neq 0$), then solve.

Claim (solution). A solution is

$$E(x) = x - 4, \qquad N(x) = (x-4)\,(2 + 3x + x^2).$$

Check quickly: if $x \neq 4$, then $E(x) \neq 0$ and $y_i = f(\alpha_i)$, so equality holds. At $x = 4$, $E(4) = 0$, so the constraint becomes $0 = N(4)$, which also holds.

Compute $N(x)$ in $\mathbb{F}_7[x]$:

$$(x-4)(x^2+3x+2) = x^3 + 3x^2 + 2x - 4x^2 - 12x - 8 = x^3 - x^2 - 10x - 8.$$

Reduce mod 7: $-1 \equiv 6$, $-10 \equiv -3 \equiv 4$, $-8 \equiv -1 \equiv 6$. Hence

$$N(x) = x^3 + 6x^2 + 4x + 6 \in \mathbb{F}_7[x].$$

Then recover

$$f(x) = \frac{N(x)}{E(x)} = \frac{x^3 + 6x^2 + 4x + 6}{x - 4} = x^2 + 3x + 2,$$

which matches the original polynomial (written in descending order). Thus we decoded uniquely and corrected the error.

Remark 7.1. Berlekamp–Welch is a purely algebraic decoding method: solve a linear system, then divide polynomials. It exemplifies the "functions evaluated at points" ideology.

8 Algebraic-geometry codes on curves

8.1 From $\mathbb{P}^1$ to curves

Replace $\mathbb{P}^1$ by a smooth projective curve $X/\mathbb{F}_q$ of genus g. Choose distinct rational points $P_1, \dots, P_n \in X(\mathbb{F}_q)$. Let

$$D = P_1 + \cdots + P_n.$$

Choose a divisor G with $\operatorname{supp}(G) \cap \operatorname{supp}(D) = \varnothing$.

8.2 Riemann–Roch space

Definition 8.1 (Riemann–Roch space)**.**

$$L(G) = \{f \in \mathbb{F}_q(X)^\times : (f) + G \geq 0\} \cup \{0\}.$$

8.3 Evaluation code

Definition 8.2 (AG code $C_L(D, G)$)**.** Define

$$C_L(D, G) = \{(f(P_1), \dots, f(P_n)) : f \in L(G)\} \subseteq \mathbb{F}_q^n.$$

8.4 Parameters from divisors (first estimates)

Theorem 8.3 (Basic parameter bounds)**.** *Assume* $\deg(G) < n$*. Then:*

$$k = \dim_{\mathbb{F}_q} C_L(D, G) = \ell(G) - \ell(G - D),$$

and in particular (using $\ell(G - D) = 0$ *when* $\deg(G) < n$ *and* G *disjoint from* D *in many standard setups),*

$$k \geq \deg(G) + 1 - g, \qquad d \geq n - \deg(G).$$

Remark 8.4. Compare with RS on $\mathbb{P}^1$ where $g = 0$: one recovers the MDS behavior. For $g > 0$, genus is the price: it weakens the dimension bound by g.

9 Duality and differentials: where "forms" really enter

Let Ω be the sheaf of differentials on X. There is a companion construction using differential forms, producing a code often denoted $C_\Omega(D, G)$.

9.1 Residue pairing

For suitable choices of G, one has a canonical pairing between evaluations and residues:

$$\langle (f(P_i)), (\mathrm{Res}_{P_i}\omega) \rangle = \sum_{i=1}^{n} f(P_i)\mathrm{Res}_{P_i}(\omega) \in \mathbb{F}_q,$$

where ω is a differential with controlled divisor.

Theorem 9.1 (Dual code description (standard AG-code duality))**.** *Under standard hypotheses,*

$$C_L(D, G)^\perp \cong C_\Omega(D, G),$$

and parity checks may be realized by residue vectors $(\mathrm{Res}_{P_1}\omega, \ldots, \mathrm{Res}_{P_n}\omega)$*.*

Remark 9.2. This is the precise sense in which coding theory touches "differential forms": the dual constraints (parity checks) can be expressed using global properties of differentials and residues (global residue theorem, Riemann–Roch, Serre duality).

10 Goppa codes: classical polynomial definition and AG interpretation

There are several closely-related notions called "Goppa codes" in the literature; the most common in coding theory/crypto is the *classical* (alternant) Goppa code.

10.1 Classical Goppa code (polynomial presentation)

Fix:

- a field extension $\mathbb{F}_{q^m}$,
- a support set $L = \{\alpha_1, \ldots, \alpha_n\} \subseteq \mathbb{F}_{q^m}$ of distinct elements,

- a polynomial $g(x) \in \mathbb{F}_{q^m}[x]$ with $g(\alpha_i) \neq 0$ for all i.

Definition 10.1 (Classical Goppa code $\Gamma(L, g)$)**.** Define $\Gamma(L, g) \subseteq \mathbb{F}_q^n$ by

$$\Gamma(L, g) = \left\{ c = (c_1, \ldots, c_n) \in \mathbb{F}_q^n : \sum_{i=1}^{n} \frac{c_i}{x - \alpha_i} \equiv 0 \pmod{g(x)} \right\},$$

equivalently via a parity-check matrix built from evaluations of $1/g(\alpha_i)$ and powers of α_i.

Remark 10.2. This condition is a "rational function congruence" that, after linearization, becomes parity-check equations over $\mathbb{F}_q$. These codes have strong distance guarantees and efficient decoding algorithms; they are central in code-based cryptography.

10.2 AG interpretation (divisor language)

Conceptually, the fractions $\frac{1}{x-\alpha_i}$ and the modulus $g(x)$ are about controlling poles/zeros of rational functions (and/or differentials) on a curve. On $\mathbb{P}^1$, $g(x)$ defines a divisor of zeros, and the support points α_i define a divisor D. Goppa/alternant codes can be viewed as *subfield subcodes* of certain AG codes on $\mathbb{P}^1$ (or closely related constructions), explaining why they inherit strong algebraic structure while living over the smaller base field $\mathbb{F}_q$.

11 A road map for further study

11.1 Combinatorics / information theory

- Gilbert–Varshamov bound (existence of good codes).
- Asymptotics: rate vs relative distance; entropy estimates of Hamming volumes.

11.2 Algebra / algorithms

- Syndrome decoding, cosets, standard array.
- BCH codes, minimal polynomials, cyclotomic cosets.
- Berlekamp–Massey, Euclidean algorithm decoding, key equations.

11.3 Algebraic geometry

- Riemann–Roch and explicit curves with many $\mathbb{F}_q$-points.
- Serre duality and the dual AG code via differentials/residues.
- Towers of curves and asymptotically good AG codes.

Exercises (selected)

1. Prove the distance statements for the parity-check code and the Hamming $(7, 4, 3)$ code.
2. For RS over $\mathbb{F}_7$ with $n = 6, k = 3$, pick a different message polynomial and simulate 1-error decoding via Berlekamp–Welch.

3. Prove Singleton bound and show RS meets it.
4. On $\mathbb{P}^1$, interpret RS as $C_L(D, G)$ with $G = (k-1) \cdot \infty$. Compute $\ell(G)$ directly.
5. Read a proof of AG duality $C_L(D, G)^\perp \cong C_\Omega(D, G)$ and identify where residues enter.